

Subcortical Aphasia

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We critically review the literature on subcortical aphasia, suggest that a number of traditional concepts regarding mechanisms of aphasia are inconsistent with now abundant data, and propose several new hypotheses. The absence of aphasia in 17 reported cases of dominant hemisphere striatocapsular infarction and the finding of nearly every conceivable pattern of language impairment in 33 different reported cases of striatocapsular infarction provide strong evidence against a major direct role of the basal ganglia in language and against disconnection or diaschisis as mechanisms of nonthalamic subcortical aphasia. However, detailed consideration of the vascular events leading to striatocapsular infarction strongly suggests that associated linguistic deficits are predominantly related to sustained cortical hypoperfusion and infarction not visible on structural imaging studies. Thalamic disconnection, as may occur with striatocapsular infarcts with extension to the temporal stem and putamenal hemorrhages, may also contribute to the language deficits in some patients. Review of the literature on thalamic infarction, in conjunction with previously unreported anatomic details of four cases, suggests that what infarcts in the tuberothalamic artery territory and the occasional infarcts in the paramedian artery territory associated with aphasia have in common is damage to the frontal lobe–inferior thalamic peduncle–nucleus reticularis–center median system that may be involved in regulating the thalamic gate in attentional processes. Disruption of attentional gating in the pulvinar and lateral posterior nuclei resulting from such

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lesions may impair selection of specific neuronal networks in the projection field of these nuclei that serve as the substrate for lexical–semantic function, which is in effect a disruption of a type of working memory, as defined by Goldman–Rakic. We define this as a defect of selective engagement. © 1997 Academic Press

Since the advent of computed tomographic imaging techniques in the 1970s, there have been numerous studies documenting aphasia after subcortical lesions, and a number of theories have been proposed to account for the language phenomena observed in these studies. Alexander, Naeser, and Palumbo (1987) suggested that the basal ganglia are not involved in language, but that certain deep white matter pathways may be important. Crosson (1985, 1992) suggested that the basal ganglia regulate the release of cortically generated language segments, but do not participate in the actual formulation of such segments. Wallesch and Papagno (1988) suggested that the basal ganglia participate in the selection of the most appropriate of multiple cortically generated lexical alternatives.

A variety of functions also have been suggested for the thalamus. Ojemann (1983) and others suggested that the thalamus is involved in attentional mechanisms subserving language. Penfield and Roberts (1959) thought that the thalamus was involved in integrative language functions. Schaltenbrand (1975) thought that the thalamus controlled the release and inhibition of preformed speech patterns (similar to Crosson's theory for the basal ganglia). Crosson (1985, 1992) suggested that the thalamus, in addition to being involved in basal ganglia regulatory activity, was involved in semantic feedback between language formulation and language decoding centers.

Problems with these theoretical analyses of basal ganglia and thalamic function prevent us from endorsing any one. In particular, in reports of aphasia due to basal ganglia and internal capsule lesions, critical analysis of circulatory dynamics has generally been lacking. Such an analysis is conducted below and leads to conclusions significantly different from those drawn by the above theorists. Regarding the thalamus, whereas we generally endorse the attentional concept espoused by Ojemann (1983), we will focus upon physiological properties of thalamic neurons and the role of the nucleus reticularis. We will begin with a critical review of nonthalamic subcortical aphasia, then consider thalamic aphasia, and conclude with some hypotheses regarding the role of the thalamus and basal ganglia in cortical function. Many of our ideas, particularly those on thalamic aphasia, will be stated with a degree of finality not at all justified by the limited data available. We have done this deliberately in the interests of clarity and brevity and to provoke further research.

NONTHALAMIC SUBCORTICAL APHASIA

To date, no compelling evidence that nonthalamic subcortical structures are directly involved in language function exists. In the following para-

graphs, we will develop two conclusions regarding nonthalamic subcortical aphasia. The first is that nonthalamic subcortical aphasia due to ischemic lesions mainly results from some combination of sustained cortical hypoperfusion (Skyhøj Olsen, Bruhn, & Öberg, 1986), cortical infarction not detected by imaging studies (Lassen, Skyhøj-Olsen, Hajgaard, & Shriver, 1983; Weiller, Willmes, Reiche, Thron, Insensee, Buell, & Ringelstein, 1993), and cortico-thalamic disconnection. Our second conclusion, closely related, is that nonthalamic hemorrhagic lesions, if small or moderate in size (e.g., Alexander & Lo Verme, 1980), may produce features of cortico-thalamic disconnection, whereas large lesions produce manifestations that reflect both thalamic disconnection and compressive ischemia of the directly overlying cortex.

Four major alternative explanations have been offered to account for nonthalamic subcortical aphasia: (1) diaschisis, or physiological dysfunction of structures remote from the damaged area (Perani, Vallar, Cappa, Messa, & Fazio, 1987; Vallar, Perani, Cappa, Messa, Lenzi, & Fazio, 1988); (2) infarction of basal ganglia structures directly involved in language processes (Damasio, Damasio, Rizzo, Varney, & Gersh, 1982; Wallesch & Papagano, 1988); (3) disconnection of cortical structures involved in language (Alexander et al., 1987); and (4) deregulation of release of cortically formulated language segments into an output stream (Crosson, 1985, 1992). We will show that, with the exception of diaschisis related to thalamic disconnection, none of these explanations satisfactorily accounts for the data. We will focus on the striatocapsular infarct because it represents a relatively stereotyped subcortical lesion caused by a specific vascular event, occlusion of the proximal middle cerebral artery by thrombus or embolus. Because it is a relatively stereotyped lesion, anatomic variability can be substantially discounted as a potential explanation for variability in clinical manifestations. Striatocapsular infarcts routinely involve the head of the caudate nucleus (Cd_H), the putamen (Put), and the anterior limb of the internal capsule (IC_{AL}). There may be some extension into the external capsule, the claustrum, lateral portions of the globus pallidus, and the anterior part of the periventricular corona radiata.

Infarction of head of caudate, putamen, and anterior limb of internal capsule (striatocapsular infarction). Table 1 provides our ratings of fluency, articulation, comprehension, repetition, naming, and the presence of paraphasic errors for all case descriptions we could find of aphasic patients with striatocapsular infarcts. Impairment was rated as mild if it was detected only with formal testing or was present in subtle form during bedside clinical evaluation. It was rated as moderate if it was readily evident during bedside exam or if the report noted impairment but did not provide detail sufficient for quantification. Impairment was rated as severe if it was substantially disabling during attempts at spontaneous language or the patient was essentially unable to perform it at all (e.g., naming, repetition) during focused testing.

TABLE 1

Linguistic Features of 33 Reported Cases of Infarction of the Head of the Caudate Nucleus, the Putamen, and the Anterior Limb of the Internal Capsule (Striatocapsular Infarction) (Alexander et al., 1987; Basso, Sala, & Farabola, 1987; Cappa et al., 1983; Damasio et al., 1982; Damasio, Eslinger, & Adams, 1984; Lieberman, Ellenberg, & Restum, 1986; Mazzocchi & Vignolo, 1979; Mendez et al., 1989; Puel et al., 1984; Robin & Schienberg, 1990; Wallesch, 1985)

Impairment	Fluency	Articulation	Comprehension	Repetition	Naming	Paraphasias
None	13	1	8	16	8	8
Mild	1	3	8	3	4	4
Moderate	13	15	9	9	12	17
Severe	5	2	8	5	9	2

It can be seen from Table 1 that nearly every possible degree of impairment in these various aspects of language has been reported. Despite the fact that these are patients with very circumscribed anterior lesions, many are fluent but have moderate to severe impairment in comprehension, repetition, and naming and produce paraphasic errors, in a manner similar to patients with posterior cortical lesions resulting in Wernicke's or conduction aphasia. Others have nonfluent output, with or without impairment of comprehension. Furthermore, 17 additional patients with this lesion have been reported who had no language disorder at all (Donnan, Bladin, Berkovic, Longley, & Saling, 1991; Ichikawa & Kageyama, 1991; Mazzocchi & Vignolo, 1979; Puel, Demonet, Cardebat, Bonafé, Gazounaud, Guiraud-Chaumeil, & Rascol, 1984; Weiller, Ringelstein, Reiche, Thron, & Buell, 1990). Even if we limit our consideration to patients with Cd_H-Put-IC_{AL} lesions for whom computed tomographic (CT) scans were included in the report (Table 2), the same picture emerges. Single investigators have reported the same phenomenon. Kertesz (1992) reported that among five patients with striatocapsular infarcts, one had a Broca's, one a transcortical motor, one a jargon, and two an anomic aphasia. Weiller et al. (1993) found that only 26 (46%) of 57 consecutive patients with striatocapsular infarction presented with aphasia or neglect, and

TABLE 2

Linguistic Features of 16 Reported Cases of Infarction of the Head of the Caudate Nucleus, the Putamen, and the Anterior Limb of the Internal Capsule (Striatocapsular Infarction) with Accompanying CT Documentation (Alexander et al., 1987; Cappa et al., 1983; Damasio et al., 1982, 1984; Lieberman et al., 1986; Mazzocchi et al., 1979; Puel et al., 1984)

Impairment	Fluency	Articulation	Comprehension	Repetition	Naming	Paraphasias
None	3	4	5	6	6	3
Mild	1	0	3	1	2	4
Moderate	10	10	6	6	6	7
Severe	2	1	2	3	2	1

of the 15 patients with aphasia, it was a Broca's aphasia in 4, Wernicke's aphasia in 3, a global aphasia in 2, an anomic aphasia in 4, and unclassifiable in 2. Furthermore, to the degree that there was variability in the extent of the striatocapsular infarct, there was no correlation of pattern or size of infarct with type or severity of aphasia. In short, the data do not support the existence of a "subcortical aphasia syndrome." If aphasia in these cases was due to dysfunction in the Cd_H , Put, and IC_{AL} , a coherent syndrome should have arisen; the lack of a coherent syndrome demands an alternative explanation.

In addition, if a common anatomic substrate accounts for the aphasia in striatocapsular infarction, then abnormalities of cerebral blood flow and metabolism should be similar in all cases. However, in three studies of cerebral blood flow in patients with striatocapsular infarcts, the heterogeneity in clinical manifestations discussed above was paralleled by heterogeneity in the pattern of cortical hypoperfusion. Cases 29 and 50 of Skyhøj-Olsen et al. (1986) had similar CT-demonstrated Cd_H -Put- IC_{AL} lesions; however, case 29 had markedly reduced blood flow in the mid-frontal convexity region, supramarginal gyrus, and Wernicke's area, whereas case 50 had markedly reduced blood flow in the foot of the pre- and postcentral gyri, parietal operculum, and suprasylvian supramarginal gyrus with sparing of the frontal lobe and Wernicke's area. Cases 14 and 16 of Alexander's study (1992) had very similar striatocapsular lesions and similar abnormalities in the frontal operculum and motor strip on positron emission tomography (PET) studies, but case 14 demonstrated reduced blood flow and metabolism in the parietal lobe, whereas case 16 demonstrated these abnormalities in midfrontal cortex. Cases 5 and 6 of Démonet, Celsis, Puel, Cardebat, Marc-Vergnes, & Rascol, (1992) exhibited very minor differences in the CT images of their striatocapsular lesions, but case 5 had a Broca's aphasia in association with predominantly lateral frontal hypoperfusion on single photon emission CT images of cerebral blood flow, whereas case 6 had a conduction aphasia in association with hypoperfusion in the posterior perisylvian cortex.

We suggest that this heterogeneity in language impairment, coupled with corresponding heterogeneity in the pattern of cortical hypoperfusion, is incompatible with an explanation invoking diaschisis as the mechanism, because if diaschisis were the mechanism, a lesion confined to a given structure or set of structures should produce a consistent pattern of physiologic dysfunction in connected regions, and therefore a consistent pattern of behavioral impairment. In the realm of subcortical aphasia, the best example of such consistency is aphasia associated with a dominant thalamic lesion, especially if the lesion is hemorrhagic (Crosson, 1992).

Lesions of caudate or putamen. What of the role of Cd_H and the putamen in language? Many patients with Cd_H -Put- IC_{AL} lesions are not aphasic. Patients with infarcts limited to Cd_H may be dysarthric but they are not aphasic (Caplan, Schmahmann, Kase, Feldman, Baquis, & Greenberg, 1990; Men-

TABLE 3
Linguistic Features of Nine Reported Cases of Hemorrhage into the Head
of the Caudate Nucleus

	Fluency	Articulation	Comprehension	Repetition	Naming	Paraphasias
Pozilli et al. (1987)	1	0	0	0	0	0
	1	0	1	0	0	0
Laine & Martilla (1981)	0	0	0	0	0	3 ^a
Cambier et al. (1979)	0	0	0	0	2	2
Cappa et al. (1983)	0	0	0	0	0	0
Robin and Schien- berg (1990) ^b	0	1	1	0	1	1
Stein et al. (1984)	0	0	0	0	0	0
Puel et al. (1984)	3	0	2	1	0	2
Choi et al. (1983)	0	0	0	0	0	0

Note. 0, Normal; 1, mild impairment; 2, moderate impairment; 3, severe impairment.

^a Writing only.

^b No CTs.

dez, Adams, & Lewandowski, 1989; Richfield, Twyman, & Berent, 1987; Schiff, Alexander, Naeser, & Galaburda, 1983). Even patients with hemorrhage nearly confined to Cd_H are infrequently aphasic (Table 3). Patients with infarctions limited to the putamen have not been aphasic (Aram, Rose, Rekate, & Whitaker, 1983; Damasio et al., 1982; Tandrig & Kirshner, 1985). Hemorrhages into the putamen are frequently large, providing the basis for pressure effects upon the overlying opercular cortex and for thalamocortical disconnection. However, a number of patients with putamenal hemorrhage, usually less than 2 cm in diameter, have been described who did not have aphasia (Alexander et al., 1987; Fromm, Holland, Swindell, & Reinmuth, 1985; Metter, Jackson, Kempler, Riege, Hanson, Mazziotta, & Phelps, 1986). At most, patients with lesions confined to the dominant basal ganglia have minor naming difficulty, difficulty with initiation, or ellipsis (Alexander, 1992).

Degenerative diseases involving the basal ganglia, such as Huntington's disease and Parkinson's disease, are associated with minimal aphasia, in no way approaching the severity of language impairment seen with Cd_H-Put-IC_{AL} lesions, and to the extent that subtle language impairment may be seen with these disorders, the pathophysiologic basis of that impairment is unclear (Crosson, 1992). Thus, human lesion data do not provide support for a *direct* role of either Cd_H or the putamen in language.

Connectivity of caudate and putamen. Data on the connectivity of Cd_H and the putamen also suggest that neither of these structures plays a significant role in language. Although Cd_H receives input from parietal cortex and superior and inferior temporal gyri, its major input is from the frontal lobes and its entire output is projected via the globus pallidus and substantia nigra

pars reticulata (SN_{PR}) through the ventral anterior (VA) nucleus of the thalamus to specific loci within the frontal lobes (Alexander, DeLong, & Strick, 1986; Goldman-Rakic & Porrino, 1985). Thus, if Cd_H were directly involved in language, one might predict that the characteristics of aphasia due to a Cd_H lesion would fall within the spectrum of features of Broca's aphasia, or more likely, minor Broca's aphasia (Mohr, Pessin, Finkelstein, Funkestein, Duncan, & Davis, 1978), isolated syntactic impairment (Nadeau, 1988), adynamic aphasia (Costello & Warrington, 1989; Luria, 1984), or transcortical motor aphasia (Rubens, 1976). Cambier, Elghozi, and Strube (1979) reported a patient with a left Cd_H hemorrhage who had prelinguistic, conceptual dysfunction most consistent with a form of adynamic aphasia. Caplan et al. (1990) reported abulia with infarcts limited to Cd_H .

The putamen receives cortical projections only from premotor cortex, motor cortex, and somatosensory cortex, and it projects via globus pallidus and SN_{PR} through the ventral lateral (VL_a) nucleus of the thalamus predominantly to the supplementary motor area (mesial area 6) (Alexander et al., 1986). The major output of supplementary motor area is to motor cortex and the corticospinal tract, indicating that this structure is predominantly involved in motor activity at a relatively low level. It has been difficult to infer the function of the supplementary motor area in humans because lesions of this structure are typically associated with lesions of other structures in the anterior cerebral artery territory, including mesial prefrontal cortex, cingulate gyrus, and callosal and pericallosal white matter connections. The dominant supplementary motor area may be involved in programming movements to the extent that they occur as part of a sequence of movements making up a more complex action, particularly if the action is internally initiated (Goldberg, 1985; Watson, Fleet, Gonzales-Rothi, & Heilman, 1986). Lesions in the vicinity of the dominant supplementary motor area may be associated with impairment in fluency, but not with the spectrum of language deficits noted with Cd_H -Put-IC_{AL} lesions (Goldberg, 1985). Therefore, one would not expect the putamen to be involved in language.

Function of caudate and putamen. Data on the function of Cd_H and the putamen fail to support a major role for these structures in language. The functions of Cd_H remain largely undefined and presumably relate to the functions of prefrontal cortex from which the majority of Cd_H afferents originate. One aspect of Cd_H function may involve motoric gating, providing the basis for transforming a potential movement or sequence of movements into an actual movement in response to an external (sensory) trigger (Lidsky, Manetto, & Schneider, 1985) or an internal trigger. The function of the putamen may be most economically explained as the definition of the duration of a movement within series or combinations of movements, primarily ballistic, through definition of the magnitude of the initial acceleration (Alexander & DeLong, 1985; DeLong, Alexander, Mitchell, & Richardson, 1986; Evarts, Teravainen, & Calne, 1981; Flowers, 1975, 1976; Georgopoulos, DeLong, &

Crutcher, 1983; Hallet & Khoshbin, 1980; Horak & Anderson, 1984a, 1984b; Marsden, 1987; Mink & Thach, 1991). However, the complex cytoarchitecture and neurophysiology of the basal ganglia have defied definitive experiments and the precise function of the putamen remains controversial (Alexander & Crutcher, 1990; DeLong, 1990; Penny & Young, 1983). Nonetheless, most authorities would agree that the functions of the putamen are motoric in a narrow sense.

Subcortical white matter lesions. Finally, is it possible that subcortical white matter lesions are responsible for most of the language impairment observed with subcortical strokes, as suggested by Alexander and colleagues (Alexander et al., 1987)? There are insufficient data to completely rule out this possibility, but two points are worth noting. First, these studies have not been replicated: when Weiller et al. (1990, 1993) subtracted lesion grids of patients with striatocapsular infarcts without neuropsychological deficits from those of patients with aphasia or neglect (assessed at a time after stroke comparable to that of Alexander et al.), they found no areas of consistent involvement. Second, many of the periventricular frontal white matter lesions incriminated by Alexander et al. (1987) are seen commonly in Binswanger's disease (Ishii, Nishihara, & Imamura, 1986), which, in common with other "subcortical" dementias, is noteworthy for the absence of language impairment other than loss of fluency.

The white matter lesions thought to cause aphasia fall in the lenticulostriate endzone. The extent of lesions in this region provides a rough measure of the severity of middle cerebral artery territory ischemia (see below). It also may provide some measure of the distribution of severest cortical ischemia, as white matter in the centrum semiovale derives its blood supply both from lenticulostriate vessels and deep penetrators from cortical vessels. Thus, in nonthalamic subcortical strokes causing aphasia, it is generally not possible to separate the direct effects of the white matter lesions from the effects on the cerebral cortex of the associated vascular events.

Putamenal hemorrhages. Putamenal hemorrhages severely damage the putamen itself. However, they also disconnect the thalamus from language cortex, and they directly compress overlying perisylvian cortex. There is evidence that both of these mechanisms may account for aphasia stemming from putamenal hemorrhage. A number of patients with putamenal hemorrhage have been reported who had impaired naming and comprehension with relatively spared repetition (Alexander & Lo Verme, 1980; Glosser, Kaplan, & Lo Verme, 1982; Metter et al., 1986; Robin & Schienberg, 1990), and many of these patients produced neologisms if not extended jargon, as may be seen in thalamic aphasia. However, many patients with putamenal hemorrhage demonstrate as much or more impairment of repetition than of other aspects of linguistic function. As repetition is spared in thalamic aphasia, this strongly suggests direct compromise of perisylvian cortex (Alexander et al., 1980, 1987; Aram et al., 1983; Cappa, Cavalotti, Guidotti, Pa-

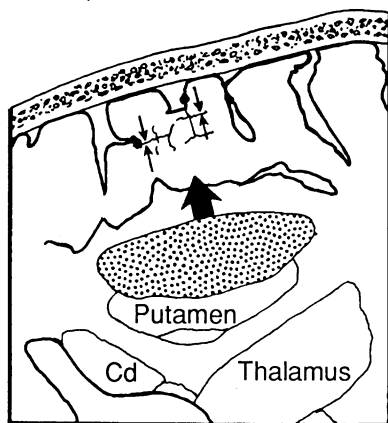


FIG. 1. Potential pressure effects of putamenal hemorrhage (stippled area). Force directed laterally, against the unmovable obstacle posed by the skull, will tend to compress small vascular branches running perpendicular to the force vector and parallel to the cortical surface. This effect will be maximal in perisylvian cortex, which immediately overlies the putamen.

pagno, Vignolo, 1983; Glosser et al., 1982; Metter et al., 1986; Naeser, Alexander, Helm-Estabrooks, Levine, Laughlin, & Geschwind, 1982; Puel et al., 1984). This suggestion is supported by blood flow studies that demonstrate profound reduction in cortical blood flow (14–25 ml/100 g/min, compared with normal values of 50–60 ml/100 g/min) throughout language cortex, including the perisylvian region, in association with putamenal hemorrhage (Skyhøj Olsen et al., 1986), in marked contrast to the far more subtle abnormalities in cortical blood flow routinely observed with thalamic lesions (Baron, D'Antora, Serdaru, Pantano, Bousser, & Samson, 1986; Caselli, Graff-Radford, & Rezai, 1991). Cortical blood flow reductions of the magnitude seen with putamenal hemorrhage are almost certainly due to compression-related ischemia, rather than cortical hypofunction caused by disconnection (Skyhøj Olsen et al., 1986). We suggest that the perisylvian language dysfunction (and perhaps components of nonperisylvian language dysfunction) may be related to compression of the directly overlying posterior perisylvian cortex by the putamenal mass; this may produce ischemia in the distribution of small vessels traveling parallel to the skull for which the vector of the compressive force is directed at 90° to the axis of the vessel and is potentially most capable of collapsing the vessel (Fig. 1). This mechanism has been invoked to account for the pattern of spinal cord damage observed in patients with compressive cervical myelopathy due to rheumatoid arthritis (Hughes, 1977; Nakano, Schoene, Baker, & Dawson, 1978). Insufficient data are provided in the literature regarding size and precise locus of hematoma, or degree of cortical atrophy and ventricular enlargement (which might mitigate compressive effects) to enable us to test this hypothe-

sis. Sufficiently detailed postmortem studies are unavailable. The compressive ischemia of the cortex may be sufficient to produce microinfarction, neuronal dropout, or merely transient neurophysiologic dysfunction.

Cortical ischemia as a major contributor to nonthalamic subcortical aphasia. Our argument here requires some review of the hemispheric vascular anatomy (Ghika, Bogousslavsky, & Crutcher, 1990). The cortical and subcortical regions potentially involved in linguistic function that are the subject of this discussion are almost always supplied by the middle cerebral artery or its branches (Beevor, 1909; Van der Zwan & Hillen, 1991; Zeal & Rhoton, 1978), with the notable exceptions of the thalamus, which is supplied by branches of the posterior cerebral arteries, posterior communicating arteries, and the basilar artery tip (Saeki & Rhoton, 1977; Zeal & Rhoton, 1978); the posterior limb of the internal capsule, which is supplied by branches of the anterior choroidal artery; and the genu of the internal capsule, which is usually supplied by a twig from the carotid artery (Ghika et al., 1990). The head of caudate nucleus, the putamen, and IC_{AL} are supplied by lenticulostriate arteries, all but the most ventromedial of which are branches of the proximal or M1 portion of the middle cerebral artery, which extends from the carotid terminus to the middle cerebral artery bifurcation.

Striatocapsular (Cd_H–Put–IC_{AL}) infarcts and nonlacunar infarcts of the white matter in the deep centrum semiovale and periventricular regions are caused by occlusion of the middle cerebral takeoff at the carotid T-junction (the point at which the internal carotid artery bifurcates to form the anterior and middle cerebral arteries); the M1 portion of the middle cerebral artery; or, less often, the internal carotid artery in conjunction with poor circle of Willis collateralization (Fig. 2). This may be due to cardiogenic or artery to artery thromboembolism or in situ thrombosis (Adams, Damasio, Putman, & Damasio, 1983; Bladin & Berkovic, 1984; Bozzao, Fantozzi, Bastianello, Bozzao, & Fieschi, 1989; Donnan et al., 1991; Levine, Lagreze, Dobkin, & Tursk, 1988; Ringelstein, Biniek, Weiller, Ammeling, Nolte, & Thron, 1992; Santamaria, Graus, Rubio, Arbizu, & Peres, 1983; Weiller et al., 1990, 1993; Yamauchi, Fukuyama, Yamaguchi, Miyoshi, Kimura, & Konishi, 1991). These events have the effect of eliminating or severely reducing middle cerebral artery derived blood flow to the cortex. The vascular territory of the middle cerebral artery then becomes supplied predominantly or entirely by end-to-end cortical anastomoses between middle cerebral artery branches, anterior cerebral artery branches, and posterior cerebral artery branches. The very lowest perfusion pressure occurs in the endzone of the lenticulostriate vessels in the centrum semiovale and periventricular region, particularly when there is T-junction or M1 occlusion. It is likely that the size of the deep white matter infarction provides a gross measure of the degree of middle cerebral artery territory ischemia caused by the T-junction occlusion (Højer-Pedersen & Petersen, 1989; Yamauchi et al., 1991). Very low perfusion pressure is also likely to be present in the insula and perisylvian cortex,

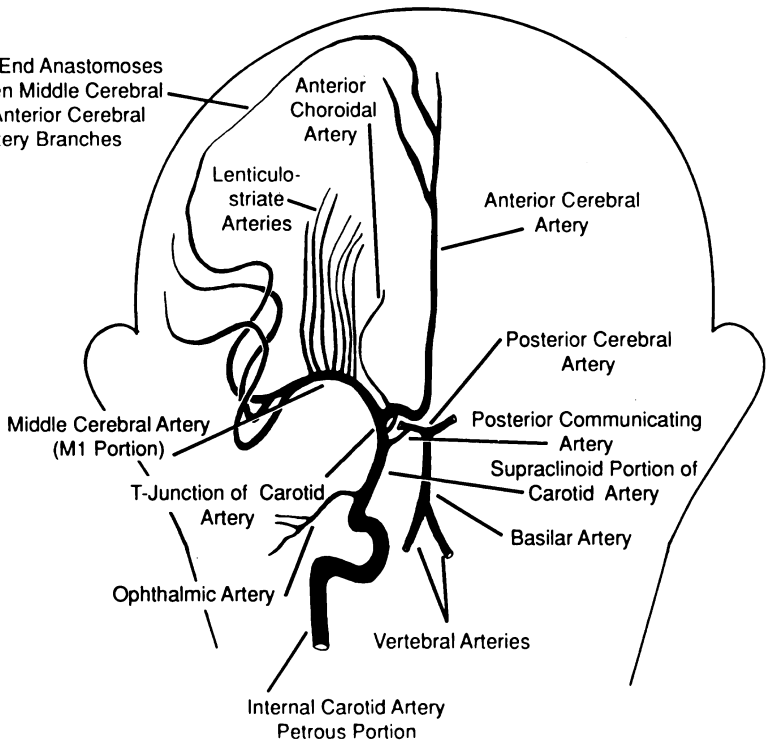


FIG. 2. Semischematic antero-posterior view of cerebral vascular anatomy relevant to the problem of nonthalamic subcortical aphasia.

the cortical regions maximally distant from collateral supply from the anterior and posterior cerebral arteries. In the case of M1 occlusion, the orifice of multiple lenticulostriate vessels may be directly occluded. As these vessels have essentially no collaterals, there follows an extensive infarction in their territory (the typical striatocapsular infarct) (Ringelstein et al., 1992).

With either T-junction or M1 occlusion, functional and anatomic sparing of the cerebral cortex will depend on the duration of proximal middle cerebral artery occlusion and the adequacy of cortical collaterals (Bozzao et al., 1989; Ringelstein et al., 1992; Weiller et al., 1993), which are highly variable in size within a given brain and between different individuals (Vander Eecken & Adams, 1953). In some cases, there will be some residual flow through the M1 that may assist in cortical sparing (Bladin & Berkovic, 1984). In the presence of poor collateralization with cortical blood flow of 10 mg/100 g/min or less, cystic infarction of some or all of the cortex supplied by the middle cerebral artery will occur (Bozzao et al., 1989; Skyhøj Olsen, Friberg, & Lassen, 1987). Lesser degrees of ischemia, associated with some-

what better collateral supply to some or all of the middle cerebral artery cortical branches, may result in ischemic neuronal dropout (Lassen et al., 1983; Van Buren & Borke, 1969), transient neuronal dysfunction associated with blood flow in the range of 18 to 23 ml/100 g/min (Heiss, 1992; Hossman, 1994; Skyhøj Olsen et al., 1986), normal neuronal function associated with an increased oxygen fraction (blood flow reduced but greater than 23 ml/100 g/min), or a normal oxygen extraction fraction and blood flow but increased cerebral blood volume in association with vasodilation (Powers, 1991). Cortical ischemia or infarction may also occur because of distal embolization from clot at the T-junction or M1.

Normal cortical blood flow is in the 50–60 ml/100 g/min range. In aphasic patients with infarcts associated with middle cerebral artery occlusion that are limited to the striatocapsular region or the periventricular white matter, cortical blood flow is consistently below 25 ml/100 g/min, as measured by the two-dimensional ^{133}Xe technique, which tends to overestimate flow within lesions because of Compton scatter; these studies also reveal impairment of cerebrovascular autoregulation in the involved cortex (Skyhøj Olsen et al., 1986, 1987). Blood flow this low, with impairment of autoregulation as well, strongly suggests that associated cerebral dysfunction is secondary to reduced blood flow rather than reduced cerebral blood flow being caused by cortical deafferentiation or dysfunction (Skyhøj Olsen et al., 1987). However, cortical blood flow of this magnitude is also sufficient to prevent gross infarction of cortical tissue (Hossman, 1994; Skyhøj Olsen, Larsen, Herning, Skriver, & Lassen, 1983). Patients with striatocapsular infarcts and persistent, poorly collateralized, proximal middle cerebral artery occlusion are highly likely to demonstrate persistent aphasia or neglect and severely reduced cortical blood flow, whereas patients with striatocapsular infarcts associated with M1 stenosis or early recanalization of the middle cerebral artery are much less likely to demonstrate persistent behavioral deficits and tend to have normal cortical blood flow (Weiller et al., 1990). For example, Weiller et al. (1993) reported that among 6 adequately studied patients with striatocapsular infarcts *without* aphasia or neglect, cortical end-to-end collaterals were good in 5 and moderate in 1, and the median duration of middle cerebral artery occlusion was 4 hr (range 1–18 hr). In contrast, among 10 patients with striatocapsular infarction *with* aphasia or neglect, cortical collaterals were moderate in 3 and poor in 4 (and indeterminate in 3), and the median duration of middle cerebral artery occlusion was 18 hr (range 4.5 hr → 10 days). Ischemic damage to the cerebral cortex may be undetected by CT or magnetic resonance imaging (MRI) in such cases because these imaging modalities have a sensitivity of only 70 or 80% for such lesions (Campbell, Hauser, Stevens, Wahner, Baker, & Folger, 1978; Kertesz, Black, Nicholson, & Carr, 1987; Nadeau, Jordan, & Mishra, 1993). On the other hand, Weiller et al. (1993) reported that of 14 patients with striatocapsular infarcts and aphasia or neglect, MRI scans repeated 1 year later revealed patchy areas

of infarction in the insular cortex in 4 patients and marked focal cortical atrophy in 9 patients. These data indicate that delaying MRI may enhance its sensitivity for ischemic cortical lesions, and they provide strong support for the idea that aphasia in patients with striatocapsular infarcts is due to cortical damage and not disconnection, diaschisis, or dysfunction of subcortical structures.

This concept of frequent, widespread, radiographically undetectable cerebral damage might seem to conflict with pathologic studies that show cortical infarcts to be rather sharply circumscribed with an abrupt transition between normal and abnormal tissue (Nedergaard, Astrup, & Klinken, 1984). However, studies like this have included small numbers of patients, and may not have explored the full spectrum of lesions (see Van Buren & Borke, 1969). Animal studies suggest that although prolonged or permanent proximal middle cerebral artery occlusion leads to extensive, confluent, and sharply marginated cortical infarctions, briefer proximal middle cerebral artery occlusion leads to selective necrosis characterized by piecemeal disappearance of cell bodies of cortical neurons with relative preservation of the integrity of the tissue framework and the presence of intervening areas of apparently normal tissue (DeGirolami, Crowell, & Marcoux, 1984). It may also be that more subtle degrees of ischemic damage manifest only as attenuation of axo-dendritic structure, as occurs in Alzheimer's disease (Terry, Mosliah, Salmon, Butters, DeTeresa, Hill, Hansen, & Katzman, 1991). The maintenance of normal axo-dendritic structure and function requires protein synthesis, which occurs in the cell body and is completely suppressed at blood flow rates below 35 ml/100 g/min. It also requires axoplasmic flow, which is critically dependent on energy metabolism, which becomes impaired at flow rates below 26 ml/100 g/min (Hossman, 1994). The impact of other destructive processes associated with ischemia, such as glutamate-mediated excitotoxicity, may also be maximal in the vicinity of axons and dendrites. Thus, immunohistochemical studies of synaptophysin, a marker for synapses, may be a more informative way of ascertaining the full ramifications of ischemic injury.

Thus, we endorse the conclusion of Skyhøj Olsen et al. (1986) and Weiller et al. (1993) that most of the language dysfunction in nonthalamic subcortical aphasia is related to primary decreases in cortical perfusion. The variability in the adequacy of distal (cortical) anastomoses with individual middle cerebral cortical branches, the variability in the cortical vascular distribution of emboli from T-junction or M1 thrombus or embolus, and the variable occurrence and timing of M1 or T-junction recanalization offer plausible explanations for the variability in the clinical presentation of Cd_H -Put- IC_{AL} infarcts, in contrast to such alternative explanations as diaschisis, damage to basal ganglia structures directly involved in language function, or deep white matter disconnection. On the other hand, we cannot exclude the possibility that these factors have obscured more subtle contributions of the basal ganglia

TABLE 4

Linguistic Features of Seven Reported Cases of Infarction of the Head of the Caudate Nucleus, the Putamen, and the Anterior Limb of the Internal Capsule (Striatocapsular Infarction) with Extension to the Temporal Stem

	Fluency	Articulation	Comprehension	Repetition	Naming	Paraphasias
Barat et al. (1981)	0	0	1	2	1	2
Alexander et al. (1987)	3	3	3	3	3	
	3	3	3	3	3	
	3	3	3	3	3	
	3	3	3	3	3	
Naeser et al. (1982)	0	0	2	2	3	2
	0		2	2	2	1

Note. 0, Normal; 1, mild impairment; 2, moderate impairment; 3, severe impairment.

to language function, including possible roles in segmentation and sequencing (K. M. Heilman, personal communication) or in maintaining working memory (M. P. Alexander, personal communication).

Cd_H–Put–IC_{AL} infarcts with extension to the temporal stem. To this point we have argued that the linguistic manifestations of nonthalamic subcortical lesions are attributable entirely to variable cortical damage or dysfunction that is not apparent on CT or MRI. If extent of cortical damage is the only operative factor, and if there is no correlation between the extent of cortical and subcortical lesions, then the extent of the subcortical lesion should be immaterial. Eighty percent of lenticulostriate vessels originate from middle cerebral artery before it breaks into two or more trunks (i.e., the M1 portion) (Umansky, Gomes, Dijovny, Diaz, Ausman, Mirchandani, & Berman, 1985). Therefore, at least 80% of the time *only* the adequacy of cortical anastomoses should determine the loci of cortical damage, and only 20% of the time is there any potential for preponderant ischemia in cortical vessels immediately overlying subcortical infarction, as thrombus extends into a single middle cerebral artery branch supplying both a portion of cortex and underlying subcortical structures. That is, 80% of the time there is no basis in middle cerebral artery structure for any correlation at all between the subcortical extent of the lesion and the cortical extent of the lesion. However, comparison of Table 4 with Table 1 suggests that patients with Cd_H–Put–IC_{AL} lesions extending to involve the temporal stem may have greater impairment of comprehension, repetition, and naming than patients without such extension; i.e., posterior extension of the subcortical lesion is associated with linguistic features suggesting posterior extension of the cortical lesion. Classifying impairment in these three aspects of language function as none or mild versus moderate or severe, there is no significant difference in comprehension or naming, but repetition is significantly worse in the temporal stem group (Yates corrected $\chi^2 = 5.542, p = .019$).

How can this be explained? The temporal stem is supplied by the anterior

choroidal artery, which also supplies most of the posterior limb of the internal capsule (Ghika et al., 1990). A temporal stem lesion might therefore be associated with ischemia in the posterior limb of the internal capsule. Lesions encompassing the entire posterior limb of the internal capsule have the effect of disconnecting the thalamus from the cortex and produce the features of thalamic aphasia (Elghozi, Strube, Signoret, Cambier, & Lhermitte, 1978; Lhermitte, 1984; McFarling, Rothi, & Heilman, 1982). More lateral subcortical lesions interrupting connections between the cortex and the thalamus would be expected to have the same effect. Thalamic disconnection could account for naming and some comprehension deficits found with Cd_H -Put- IC_{AL} lesions extending posteriorly (and studies of larger numbers of patients might bear this out), but it is unlikely to account for worse repetition, as repetition is consistently spared in thalamic aphasia.

On the other hand, striatocapsular infarction in conjunction with infarction in the territory of the anterior choroidal artery strongly suggests the presence of thrombus not only in the proximal middle cerebral artery, but also in the supraclinoid carotid artery, from which the anterior choroidal artery originates (Fig. 2). Thrombus in the supraclinoid carotid artery would also occlude the orifice of the posterior communicating artery, or the posterior cerebral artery in that 12% of patients in whom the left posterior cerebral artery is a branch of the carotid artery (Saeki & Rhoton, 1977). Either event, but particularly the latter, would compromise the ability of the posterior cerebral artery to provide collateral blood supply to the proximally occluded middle cerebral artery, and thus increase the degree of ischemia in the cortex, including the perisylvian region important for repetition.

In summary, in cases in which subcortical infarction involves both the striatocapsular region and the temporal stem (or white matter just lateral to it), there is reason to expect more severe aphasia than in isolated striatocapsular infarction. This is because the temporal stem lesion disconnects the thalamus from the cortex (aggravating naming and comprehension deficits) and because temporal stem involvement implies more severe cortical hypoperfusion due to loss of collateral supply as a result of carotid siphon thrombosis and associated posterior communicating artery occlusion. As noted above, such cortical ischemia may not be sufficiently severe to cause a lesion visible on structural imaging studies. Available data tend to support this hypothesis, at least insofar as repetition is concerned.

THE RELATIONSHIP OF THALAMIC APHASIA TO DISRUPTION OF THE FRONTAL-INTRALAMINAR AXIS

Our third major conclusion, after those we drew on the origin of nonthalamic subcortical aphasia (see above), is that the necessary and sufficient lesion for the production of thalamic aphasia either involves the pulvinar-lateral posterior (LP) complex, which is extensively connected with tempo-

FRONTAL CORTEX

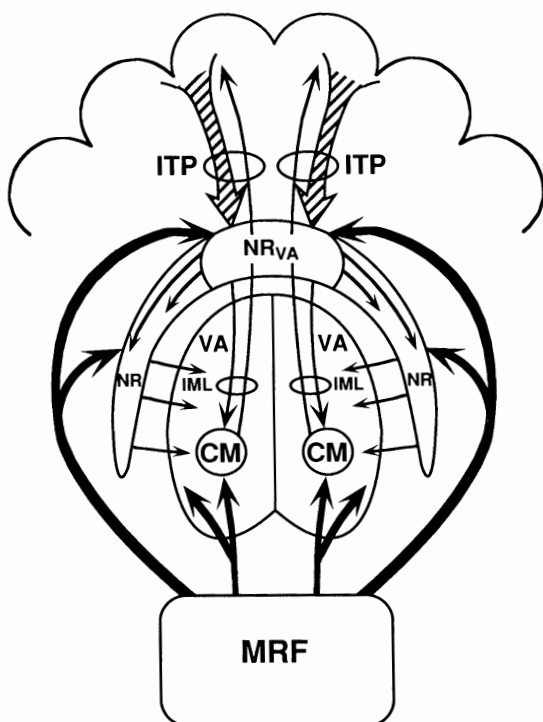


FIG. 3. Schematic diagram of the essential relationships in the frontal-ITP-NR-CM system. See text for details. IML, internal medullary lamina. Diffuse projections from the cerebral cortex, including the frontal lobes, to the thalamus, including CM, and projections from thalamic nuclei to the cerebral cortex, together with their collateral projections to NR, have not been diagrammed for sake of clarity.

ral, parietal, and probably frontal cortex (Asanuma, Andersen, & Cowan, 1985; Goldman-Rakic & Porrino, 1985; Jones, 1985; Van Buren & Borke, 1969), or it disrupts the system by which the frontal lobes influence nucleus reticularis thalami (NR).

Nucleus reticularis consists of a thin sheet of GABAergic neurons that envelopes the thalamus and projects to all nuclei except the anterior nuclear group. Cells in NR project exclusively back on the thalamus, synapsing on both thalamocortical neurons and inhibitory interneurons in the thalamus, particularly in the intralaminar nuclei (Steriade, Parent, Hada, 1984) (Fig. 3). Corticothalamic and thalamocortical projections synapse upon cells in NR as they travel through it. The connections between the frontal lobes and both NR and the intralaminar nuclei traverse the inferior thalamic peduncle

(ITP), which is located at the junction between the most posterior portions of IC_{AL} and the anteroventral aspect of the thalamus. The interface between the ITP and NR is located in that portion of NR, NR_{VA}, immediately anterior and ventral to the ventral anterior nucleus (VA) of the thalamus (Campos-Ortega & Cluver, 1969). A particularly dense neuropil is located in this region of NR that is the recipient of some if not all descending ITP fibers from the frontal cortex, as well as ascending fibers from the midbrain reticular formation (MRF); interconnections are then made between cells in NR_{VA} and more posterior NR cells, apparently via long dendritic bundles (Scheibel & Scheibel, 1967, 1972). Thus, the anatomic relationship between the inhibitory neurons in NR and the neurons within the thalamus, together with the connections between the frontal lobes and NR via the ITP, provide a potential means by which the frontal lobes might influence the gating of thalamic transmission to the cortex, presumably including pulvinar–LP transmission to temporal and parietal cortex.

Three sources of data suggest that the intralaminar nuclei (center median (CM) and parafascicularis (PF)) are also an intrinsic part of this frontal–NR system (Skinner & Yingling, 1977). These include the particularly extensive connectivity between the intralaminar nuclei and both NR and the frontal lobes (Steriade et al., 1984); the precisely opposite effects of low frequency intralaminar stimulation and ITP disruption on a variety of electrophysiologic phenomena (recruiting responses and slow potential shifts in frontal cortex, and sensory evoked potentials); and the fact that functional disruption of the ITP prevents driving of NR neurons by intralaminar stimulation. We will focus on CM because of its large size in humans, its relatively lateral location, and because it is the recipient of projections from the full mediolateral extent of NR_{VA} (Steriade et al., 1984). Later in the paper we will consider in more detail the functional interaction between the components of the frontal–ITP–NR–CM system. For now we will consider clinical and anatomic evidence bearing on the conclusion we drew at the beginning of this section.

Anatomic data from patients with thalamic aphasia. We will focus primarily upon patients with ischemic thalamic lesions because of the difficulty in delineating the extent of compressive and irritative effects of thalamic hemorrhages. Thalamic aphasia associated with ischemic lesions occurs primarily with lacunar infarcts in the territory of the tuberothalamic artery and less often with lacunar infarcts in the territory of the paramedian artery (in which case infarcts are frequently bilateral) (Archer, Ilinsky, Goldfader, & Smith, 1981; Barbizet, Degos, Louarn, Nguyen, & Mas, 1981; Biller, Merchant, & Emanuele, 1984; Bogousslavsky, Miklossy, Deruaz, Regli, & Assal, 1986a; Bogousslavsky, Regli, & Assal, 1996b; Bruyn, 1989; Caselli et al., 1991; Cohen, Gelfer, & Sweet, 1980; Davous, Bianco, Duval-Lota, de Recondo, Vedrenne, & Rondot, 1984; Dubois, Pillon, De Saxce, Lhermitte, & Agid, 1986; Elghozi et al., 1978; Fasanaro, Spitaleri, Valiani, Postiglione, Soricelli, Mansi, & Grossi, 1987; Fensore, Lazzarino, Nappo, & Nicolai,

1988; Gentilini, De Renzi, & Crisi, 1987; Geny, Youlis, Azoulay, Brugieres, Saint-Val, & Degos, 1991; Goldenberg, Wimmer, & Maly, 1983; Gorelick, Hier, Benevento, Levitt, & Tan, 1984; Graff-Radford, Eslinger, Damasio, & Yamada, 1984; Graff-Radford, Tranel, VanHoesen, & Brandt, 1985; Guberman & Stuss, 1983; Katz, Alexander, & Mandell, 1987; Kritchevsky, Graff-Radford, & Damasio, 1987; Lang, Feistel, Romstock, 1988; Lazzarino, Nicolai, Valassi, & Biasizzo, 1991; Lhermitte, 1984; Mazaux & Orgogozo, 1982; McFarling et al., 1982; Molnar, 1959; Mori, Yamadori, & Mitani, 1986; Puel, Demonet, Cardebat, Berry, & Celsis, 1992; Robin & Schienberg, 1990; Rousseaux, Cabaret, Lesoin, Devos, Dubois, & Petit, 1986; Sandson, Daffner, Carvalho, & Mesulam, 1991; Schott, Mauguière, Laurent, Serclerat, & Fischer, 1980; Stuss, Guberman, Nelson, & Larochelle, 1988; Tuszynski & Petit, 1988; von Cramon, Hebel, & Schuri, 1985). The tuberothalamic artery supplies primarily VA and the ventrolateral nucleus (VL) as well as ventral thalamic structures beneath these nuclei. The paramedian artery supplies the midline intralaminar nuclei, ventral portions of the dorsomedial nucleus, usually the mammillothalamic tract, and midline ventral thalamic structures (see Appendix for diagram of the thalamus, the connectivity of the major thalamic nuclei, and their vascular supply).

Before considering the anatomy of thalamic aphasia in more detail, it is worth asking what portions of the thalamus might logically be involved in language. The most obvious candidates are the pulvinar and LP nuclei, by virtue of their extensive connections with temporal and parietal regions subserving language function (Jones, 1985; Van Buren & Borke, 1969). Ischemic lesions of pulvinar and LP are very unusual, possibly because these nuclei are supplied predominantly by the posterolateral choroidal artery, which receives distal collateral supply from the posteromedial and anterior choroidal arteries (Lhermitte, 1984). Only three cases have been reported. Bogousslavsky, Regli, and Uske (1988) reported "mild dysphasia" in a patient with a small, laterally placed pulvinar lesion. Rousseaux et al. (1986) (case 6) described anomia and reduced verbal fluency with spared repetition, naming, and comprehension in a patient said, on the basis of the CT scan, to have a pulvinar lesion due to posterolateral choroidal territory infarction; some involvement of CM seems plausible. Tuszynski and Petit (1988) described mild impairment of comprehension and naming, mildly impaired fluency, occasional phonemic and semantic paraphasias, rare jargon, and normal repetition in a patient found at postmortem to have an ischemic lesion of the pulvinar (and possibly CM). Cappa, Papagno, Vallar, and Vignolo, (1986) reported five patients with hemorrhages largely dorsal to the pulvinar (but probably at least compressing it from above, and in two cases extending into it) who had very minimal language impairment. Most patients experience some anomia during electrical stimulation of the left pulvinar (Fedio & Van Buren, 1975). Unfortunately, the precise locus of stimulation responsible for the anomia is open to question because of variability in the size of

the human thalamus (Van Buren & Borke, 1972), the distance of current spread, which with the large currents used (up to 15 mA) could be as much as several millimeters (Ranck, 1975), and the potential for excitation of fibers *en passage*. Ojemann has in fact suggested that anomia and verbal memory phenomena associated with VA stimulation may be related to stimulation of the intralaminar–NR system (Ojemann, 1975). The stimulation sites in the pulvinar that elicited anomia were in the vicinity of either CM or NR (Fedio & Van Buren, 1975). Detailed behavioral studies of patients with chronic, surgically created lesions of the pulvinar have generally failed to provide evidence of aphasia (Brown, Riklan, Waltz, Jackson, & Cooper, 1971; Cooper, Riklan, & Rakic, 1974; Vilkki & Laitinen, 1976). There are several plausible reasons for this failure that are still consistent with a role by pulvinar in language. In no case was lesion location confirmed histologically. Linguistic dysfunction with pulvinar lesions may become manifest only with substantial damage, and the lesions may simply have been too small in many of the cases cited to produce a deficit, or the deficit may have been too subtle to detect except in patients in whom there was preexistent language impairment (Brown, 1974). Furthermore, the pulvinar is a large nucleus with heterogeneous connectivity that includes not just language cortex but visual, somatosensory and auditory association cortices in the temporal and parietal lobes, as well as frontal cortex. Cerebral cortical lesions producing persistent language impairment are associated with degenerative changes maximal in anterior and superior portions of the lateral pulvinar nucleus (Van Buren & Borke, 1969). It is possible that the lesions in patients without aphasia did not include this critical region. Aphasia from pulvinar lesions may also be transitory because of the existence of compensatory mechanisms. In conclusion, the role of the pulvinar in language remains insufficiently tested.

The ventral anterior nucleus, which is consistently the site of major damage with tuberothalamic infarcts, is the recipient of Cd projections via the globus pallidus and SN_{PR} and projects to prefrontal cortex. The ventrolateral nucleus (VL_a) is the recipient of putamenal projections primarily via the globus pallidus and projects almost exclusively to the supplementary motor area (Alexander et al., 1986). As we discussed in the section on nonthalamic subcortical aphasia, it is unlikely on anatomic and physiologic grounds that VA or VL are involved in language function in more than a minor capacity. The strongest challenge to this argument is posed by studies of patients, largely Parkinsonian, who have undergone stereotaxic lesions of dominant VL. Postoperative aphasia has been reported in 16 to 42%; however, persistent language impairment is relatively rare (Crosson, 1992). For the most part, the language deficits in these patients have been poorly characterized (Cooper, Riklan, Stellar, Waltz, Levita, Ribera, & Zimmerman, 1968), but Bell (1968) reported language impairment in 20% that was characterized by word finding difficulty, impaired naming to confrontation, occasional phonemic and semantic paraphasias, and reduced output with relatively spared

comprehension and repetition. Impaired object naming, category naming, word fluency, Token Test performance, Stroop performance, word pair learning, and Wechsler Adult Intelligence Scale information, similarities, vocabulary, verbal, and digit symbol scores have been reported in patients with dominant VL lesions (Almgren, Andersson, & Kullberg, 1969; McFie, 1960; Ojemann, 1976; Riklan, Levita, & Zimmerman, 1969; Shapiro, Sadowsky, Henderson, & Van Buren, 1973; Vilkki & Laitinen, 1974). Language deficits appear to have occurred in a minority of patients, and it is possible that lesions in these patients encroached upon neighboring structures. Unfortunately, clinicopathologic correlation is not available. The much lower frequency of aphasia or dysarthria with recent series using modern stereotaxic techniques suggests that lesion placement errors may well have accounted for the aphasia in these patients (Fox, Ahlskog, & Kelly, 1991). If we are correct in our assumption that a lesion of VA or VL does not account for the aphasia in patients with tuberothalamic territory infarcts, the question arises as to just what part of these infarcts is responsible for aphasia.

One approach to the problem of clinical-anatomic correlation in thalamic aphasia is to ask what tuberothalamic territory infarcts and the occasional paramedian infarcts associated with aphasia have in common. The literature presents a problem in this regard, because unless infarcts are bilateral (in which case they are reliably paramedian artery territory), tuberothalamic and paramedian territory infarcts may look similar in size and location on a single axial CT or MRI cut through mid-thalamus, and the locus of the lesion in many reports is uncertain. It is only on axial cuts through inferior thalamus that it becomes evident that paramedian territory infarcts curve medially and posteriorly (Schott et al., 1980; von Cramon et al., 1985), whereas tuberothalamic territory infarcts remain anterior and slightly lateral. The two types of infarct are therefore best distinguished radiographically at the most inferior levels, i.e., at the level of the midpoint of the anterior commissure-posterior commissure line (level H.0 in the Schaltenbrand and Bailey atlas (1959)). The two types are also distinguishable on coronal cuts: tuberothalamic infarcts are more or less cylindrical in shape, and are aligned along the rostro-caudal (z) axis; paramedian infarcts consist of an essential core immediately adjacent to the third ventricle, and to one degree or another extend laterally and rostrally in the shape of inverted wings. Finally, the two types of infarcts are generally reliably distinguished clinically, because paramedian territory infarcts are *usually* associated with acute lethargy, impairment of recent memory and frontal lobe features, but not aphasia, and they are frequently associated with disorders of eye movement because of incursion of the infarct on the third nerve nuclear complex and fascicles and structures immediately rostral to it involved in supranuclear control of eye movement (Tatemichi, Steinke, Duncan, Bello, Odel, Behrens, Hilal, & Mohr, 1992b).

To further address the question of what tuberothalamic territory and paramedian territory infarcts may have in common anatomically, we will

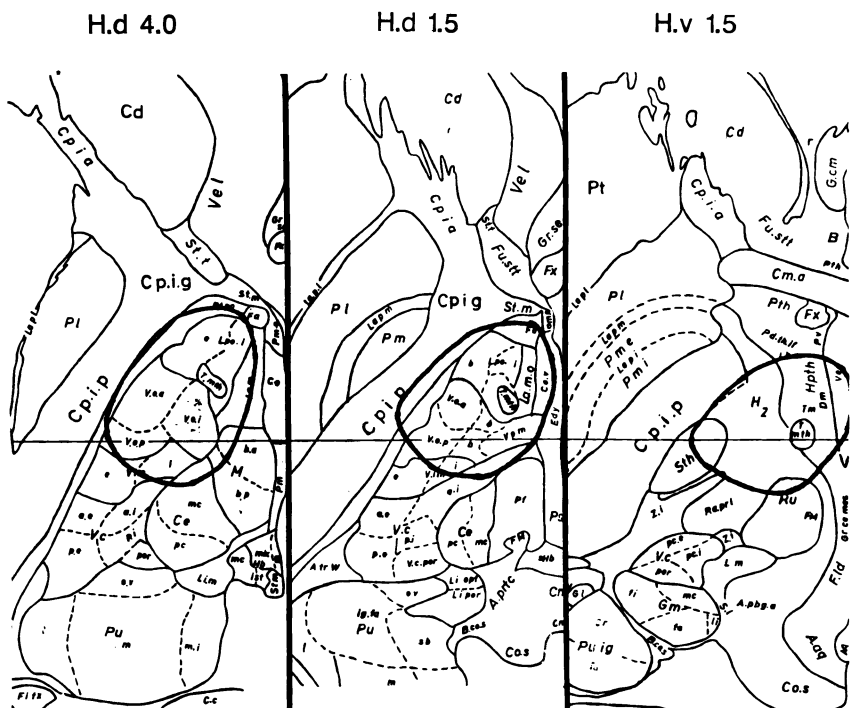


FIG. 4. Map of the thalamic lesion of case BD on axial sections (Schaltenbrand & Bailey, 1959). After identification of the appropriate planes, levels H.d 4.0 (4 mm dorsal to the anterior commissure (AC)–posterior commissure (PC) line) and H.v 1.5 (1.5 mm ventral to the AC–PC line) were plotted using a camera lucida technique. The lesion map at H.d 1.5 is an interpolation between the levels above and below. Designations use Hassler nomenclature. Note involvement at H.d 1.5 of the anteroventral margin of VA (L.po), which is the ITP–NR interface. A, AV; Ce, CM; Cp, internal capsule; D.o, L.po, VA; D.sf, LD; M, DM; Pd.th.if, ITP; Pu, Pulvinar; Rt, NR; Vc, Z.c, Z.im, VPL, VPM; V.o.a, V.o.p, Z.o, VLa; D.im, V.im, V.o.i, VLP; T.mth, mamillothalamic tract (Jones, 1990).

consider four cases, two of which we recently reported (Raymer, Moberg, Crosson, Nadeau, & Gonzalez-Rothi, 1997). All four patients had fairly typical thalamic aphasia, with anomia in spontaneous language, poor performance on word fluency tests, problems with naming to confrontation, fairly good comprehension, and normal repetition. Cases BD and CG had tuberothalamic territory infarcts. In BD, the infarct involved most of both VA (Lpo) and VL (V.im, V.o) (Fig. 4, H.d 4.0), but it extended inferiorly to involve medial and lateral portions of the ITP (Fig. 4, H.v 1.5), and in between, it involved the anteroventral portion of VA (Lpo), which is the site of the ITP–NR interface. In CG, the infarct also involved much of VL (V.im, V.o) (Fig.

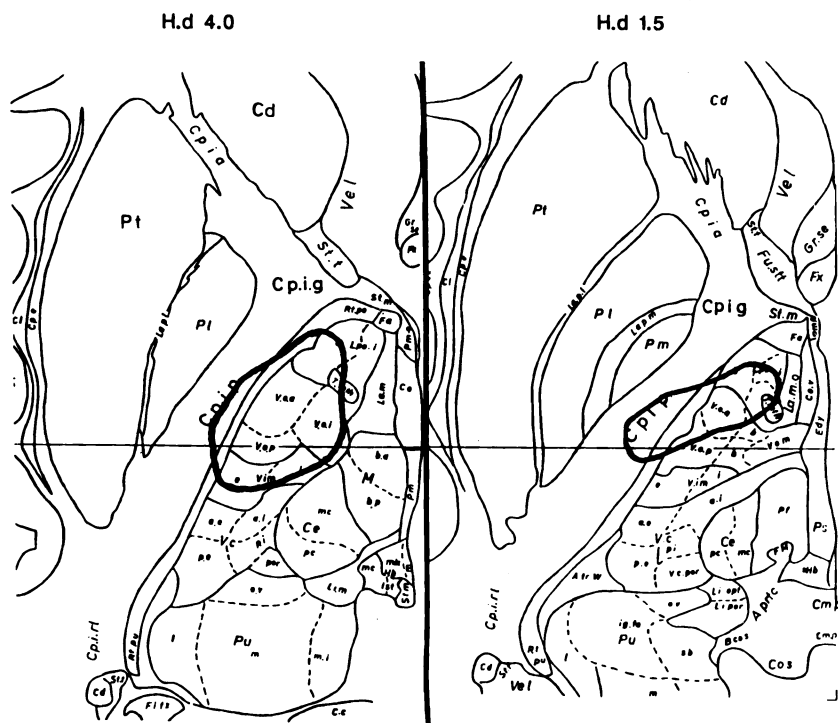


FIG. 5. Map of the thalamic lesion of case CG on axial sections (Schaltenbrand & Bailey, 1959). Mapping technique and nomenclature as described in Fig. 3. Note involvement of anteroventral aspect VA (Lpo) (the site of the ITP–NR interface) immediately lateral to the internal medullary lamina (La.m.o) (H.d 1.5).

5, H.d 4.0) and some of VA (Lpo), and although it did not extend as far inferiorly as in BD, it did involve the site of the ITP–NR interface at anteroventral VA (Fig. 5, H.d 1.5). Lesions of similar extent, associated with aphasia, have been noted by Archer et al. (1981).

We have hypothesized that damage to the ITP–NR interface might be responsible for the aphasia. However, the ITP–NR interface is usually spared in paramedian territory infarcts, which are usually more posterior. Even with those paramedian infarcts that are unusually anterior, only the most medial portions of the ITP are typically involved (Malamut, Graff-Radford, Chawluk, Grossman, & Gur, 1992). Our third case provides a possible answer to this conundrum. Patient WT experienced an unusually large paramedian territory infarct. The infarct was too posterior to involve VA or the ITP, but it included much of VL (V.im, V.o) (Fig. 6), dorsomedial nucleus (M), CM (Ce), dorsal VPL (Z.c), and LP (D.c). One might conclude in this case that either involvement of dorsomedial nucleus (DM), by virtue of its extensive

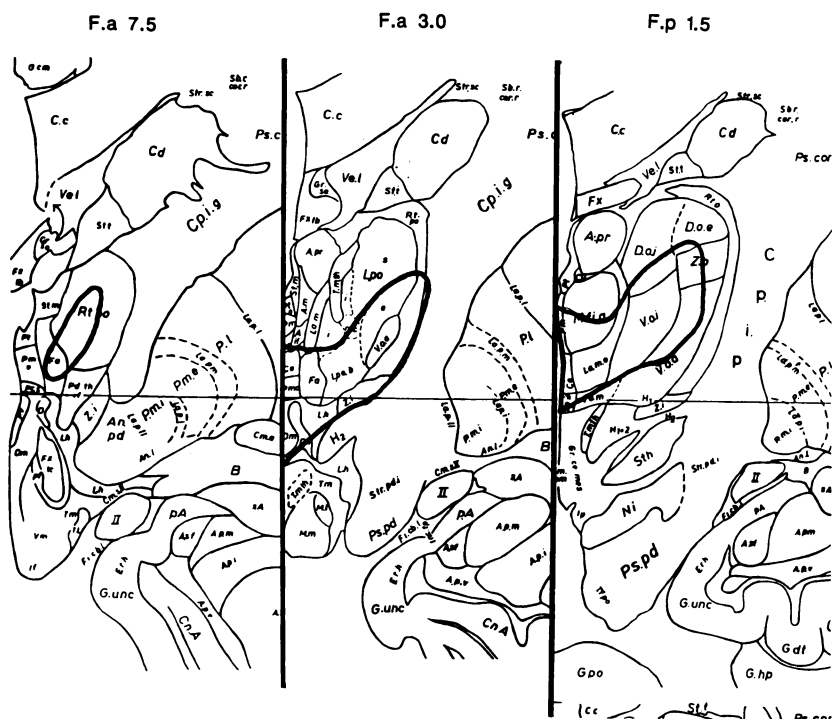


FIG. 7. Map of the thalamic lesion of case JB on coronal sections (Schaltenbrand & Bailey, 1959). Mapping technique and nomenclature as described in Fig. 3. Note involvement of Rt.po (F.a 7.5), the site of ITP–NR interface, and of the internal medullary lamina (La.m.o) immediately anterior to CM (F.p 1.5).

which a laterally extending left hemisphere paramedian infarction was not associated with aphasia are the case of Mennemier, Fennell, Valenstein, and Heilman (1992) and case 2 of Graff-Radford, Tranel, VanHoesen, and Brandt (1990), both left-handed and neither with data on language during the weeks immediately following the stroke, and case 2 of Kritchevsky et al. (1987), in which CM involvement was not reported but is conceivable given the appearance of the MRI.

Thus, we conclude on the basis of this limited evidence that the one thing that cases of tuberothalamic territory infarction and cases of paramedian territory infarction with aphasia have in common is involvement of the frontal–ITP–NR–CM system. Few published cases of tuberothalamic infarction have included sufficiently precise mapping of the lesions to determine whether NR_{VA} or projections from NR to CM are invariably involved. The main challenge to this hypothesis at present comes from reports of surgically induced thalamic lesions. Poorly characterized language dysfunction has

been reported with VL lesions carefully mapped at postmortem examination (Samra, Riklan, Levita, Zimmerman, Waltz, Bergmann, & Cooper, 1969). No language impairment has been commented upon in patients who have undergone CM thalamotomy (without pathologic confirmation of lesion locus) (Jurko & Andy, 1973, 1977).

Anatomic data from patients with lesions outside the thalamus. The concept that **thalamic aphasia is usually** due to lesions of the frontal–ITP–NR–CM system is further supported by observations of language impairment similar to that in patients with thalamic lesions in patients with infarcts in inferior portions of the genu of the IC (Tatemichi, Desmond, Prohovnik, Cross, Gropen, Mohr, & Stern, 1992a). Such lesions do not directly implicate any portion of the thalamus, but they fall directly within the ITP.

Plausibility of the frontal–ITP–NR–CM lesion as a cause of aphasia. We have provided limited anatomic evidence suggesting that lesions of the frontal–ITP–NR–CM system may provide the basis for most cases of thalamic aphasia. We now ask whether such a lesion offers a plausible explanation for aphasia.

The essential function of the thalamus is as a gated relay (Steriade, Jones, & Llinas, 1990). Thalamocortical neurons in general are highly suited, both anatomically and physiologically, for highly reliable transfer of afferent information to cortical targets. It is not known whether intralaminar nuclei operate on the same principles as other thalamic nuclei. The thalamic relay is profoundly influenced by the state of arousal of the organism, and possibly by attentional processes. Thalamic projection neurons reside in one of two states: hyperpolarized (corresponding to the slow-wave sleep state), in which case threshold for response is elevated and responses tend to be in bursts which have little relationship to afferent input, and partially depolarized (the waking state), in which case responses are single and are linearly entrained to afferent input up to 200 Hz (McCormick & Feese, 1990; Steriade et al., 1990). The state of thalamic projection neurons reflects the status of several membrane currents, which is largely determined by direct and indirect input from the cerebral cortex (McCormick & von Krosigk, 1992); direct input from several brainstem structures, particularly the midbrain and pontine reticular formation (cholinergic) (Steriade et al., 1991), locus ceruleus (noradrenergic), and the raphe nuclei (serotonergic) (McCormick & Pape, 1990); and by NR (GABAergic) (Shosaku, Kayama, Sumitomo, Sugitani, & Iwama, 1989; Steriade & Llinás, 1988). Of these many inputs, those from the MRF and NR are probably the most important. Two major systems appear to govern the activity of NR: the MRF (Steriade & Llinás, 1988) and the frontal–ITP–NR–CM system (Skinner & Yingling, 1977; Yingling & Skinner, 1977). Stimulation of the MRF, via direct projections to thalamic nuclei and via MRF projections to NR, has the effect of shifting all projection neurons in the thalamus into the partially depolarized, transmission mode. Direct noradrenergic and serotonergic input to thalamic neurons has the same effect

(McCormick & Pape, 1990). This effect is prolonged and corresponds to neurophysiologic and behavioral phenomena associated with arousal (Livingstone & Hubel, 1981).

Frontal input via the frontal-ITP-NR-CM system must have a more subtle effect on thalamic neurons, at least to the extent that it is involved in attentional processes and operates in states of wakefulness when bursting is not observed in thalamocortical neurons (Steriade, Jones, & Llinas, 1990). It has been suggested that in attentional processes, particular regions of NR may act to selectively inhibit local pool GABAergic interneurons within parts of thalamic relay nuclei, thus producing increased relay-mode firing in the relevant thalamocortical neurons (Steriade, Domich, & Oakson, 1986). Simultaneously, axon collaterals from these same regions of NR inhibit neighboring NR neurons, leading to disinhibition of GABAergic thalamic interneurons and suppression of thalamocortical neurons surrounding the relevant thalamic focus. Skinner and Yingling provided evidence that the frontal-ITP-NR-CM system is involved in this process (Skinner & Yingling, 1977; Yingling & Skinner, 1977). They found that cooling the ITP increases the amplitude of visual, auditory, and somatosensory cortical evoked potentials in a somatotopic fashion. It also interrupts the performance of goal oriented behavior, e.g., bar-press for food, presumably by preventing the selective enhancement of thalamocortical transmission necessary to pursue goal oriented activity. Stimulating the neuropil in the region of NR_{VA} suppresses evoked potentials. Skinner and Yingling (1977) suggested that "the function of the [frontal-ITP-NR-CM system] appears [to be] to keep orienting reactions under control by selectively inhibiting the ascent of sensory stimuli that are deemed by it to be irrelevant. . . . The [frontal-ITP-NR-CM system] regulates the selective and voluntary types of attention, whereas the MRF seems to control the more general and reflexive types" (pp. 54, 62). The thalamus, in conjunction with the MRF, appears to be involved in the widespread gating of sensory information to cortex that provides part of the basis for arousal, vigilance, reactive attention, and reactive behavior. The frontal-ITP-NR-CM system appears to be involved in the phasic and selective enhancement of thalamic output relevant to intentionally guided attention and behavior, at the same time that it appears to inhibit the relatively tonic and nonselective gating of sensory information mediated by the MRF-NR system.

There have since been a number of studies with results that are consistent with the pioneering work of work of Skinner and Yingling, albeit with some qualifications, and no studies that to our knowledge have fundamentally contradicted their findings and conclusions. These studies involve analysis of alterations of sensory evoked potentials and neuronal firing that occur sufficiently soon after a sensory stimulus to plausibly reflect gating of thalamocortical transmission. Studies of somatosensory evoked potentials in humans (Cheron & Borenstein, 1991; Cohen & Starr, 1987; Desmedt & Tomberg,

1989; Yamaguchi & Knight, 1990) have demonstrated changes in the amplitude of the P27 wave, thought to be generated in area 1 (which receives thalamocortical projections) (Desmedt & Tomberg, 1989), by changes in attentional set (increase), movement (marked decrease), and lesions of prefrontal cortex (marked increase). Evoked potential studies in cats have demonstrated a significant reduction of response in somatosensory cortex and CM to nonrelevant somatosensory stimuli during attention to a relevant stimulus (Ciancia, Maitte, Honoré, Lecoutre, & Coquery, 1988). Single unit studies in somatosensory cortex in monkeys (Hyvärinen, Poranen, & Jokinen, 1980) and rats (Chapin & Woodward, 1981) have demonstrated enhancement of firing in area 1 and to a much lesser extent in area 3b when stimuli are attended and a profound suppression of firing in SI (rat) in association with movement. However, there were no attention-related changes in activity in VPL (Poranen & Hyvärinen, 1982). Studies of auditory evoked potentials in humans have demonstrated enhancement of the Pa wave (latency, 22–38 msec), thought to be generated in or near auditory cortex, by attention (Hackley, Woldorff, & Hillyard, 1990; McCallum, Curry, Cooper, Pocock, & Papakostopoulos, 1983; Woldorff, Hansen, & Hillyard, 1987), and by frontal lobe lesions (Knight, Scabini, & Woods, 1989). Single unit studies in auditory cortex of monkeys have demonstrated changes in firing with latencies as short as 15–20 msec in association with attention (Benson & Hienz, 1978; Hocherman, Benson, Goldstein, Heffner, & Hienz, 1976). However, if these effects are produced at the thalamic level, they are likely to be mediated via the magnocellular medial geniculate nucleus and not via the “lemniscal” tonotopic system that traverses parvocellular medial geniculate nucleus, whose cells show no attentional effects (Weinberger & Diamond, 1987). Visual evoked potential studies in humans have not succeeded in demonstrating attentional effects on early components (i.e., before P105) (Hackley et al., 1990), nor have single unit studies in primates demonstrated any attentional modification of the firing of cells in striate cortex (Robinson, Baizer, & Dow, 1980; Wurtz & Mohler, 1976).

Unfortunately, only one of the studies mentioned above has shown that the demonstrated effects occur at the thalamic level (Ciancia et al., 1988), and it is perfectly plausible that they are primarily mediated at the cortical level. This is almost certainly the case in the visual system in primates, in which attentional effects have been demonstrated in a host of studies, but *only* in visual association cortices. The work of Weinberger and Diamond (1987) indicates that even in the cat, attentional effects at the thalamic level in the auditory system are likely to be mediated only via the magnocellular medial geniculate system. The congruence between results of single unit studies in the somatosensory system in rats (Chapin & Woodward, 1981) and evoked potential studies in humans (Cohen et al., 1987) provides some evidence of the comparability of this system in these widely disparate species. However, the apparent lack of complete agreement between the results

of Skinner and Yingling and later studies of the auditory and visual systems leads one to ask just how much the dominant attentional mechanisms in primates have shifted away from gating of thalamic sensory relay nuclei to selection processes operative in association cortices. There is good evidence that the thalamus is involved in the latter. Hemispatial neglect has been observed in patients with lesions of the thalamus (Rafal & Posner, 1987; Watson & Heilman, 1979; Watson, Valenstein, & Heilman, 1981), and neglect, extinction, and delayed reorienting of attention have been induced in monkeys through electrolytic lesions of CM-parafascicularis (Watson, Miller, & Heilman, 1978) and by injection of the GABA agonist muscimol into the pulvinar (Desimone, Wessinger, Thomas, & Schneider, 1990; Petersen, Robinson, & Morris, 1987). In a recent PET study, increased metabolic activity was demonstrated in the pulvinar in association with a task requiring identification of a letter within a surround of similar letters (LaBerge & Buchsbaum, 1990). Visual perceptive and visual constructive disorders have been documented in patients with thalamic lesions (Stuss et al., 1988).

In contrast to the extensive neurophysiologic studies of thalamic function in relation to NR, MRF, and arousal (Steriade et al., 1990), there have been essentially no neurophysiologic studies that have tested the findings and hypotheses of Skinner and Yingling regarding the frontal-ITP-NR-CM system, nor have neurophysiologic studies explored the behavior of the thalamus during attentional processes.

So far in this section we have spoken only about attention, which is the process whereby the brain regulates the locus and the breadth of its focus upon incoming sensory information. To the extent that this regulatory process involves the thalamus, it is achieved by the gating of thalamic transmission to cortex, which must have the effect of altering the state of neurons in selected regions of the cortex. If the thalamus serves to alter selectively the state of polarization, activity, or connectivity of cortical neurons in the service of attentional processes, it is plausible that it does the same thing in the service of other processes mediated by the cortex, including language (see below), that may involve in part nonsensory thalamic nuclei, such as CM-PF, pulvinar, LP, VA, VL, and DM. This service of other processes, although nonattentional, is presumably mediated by similar thalamic mechanisms. Just as attentional processes serve to focus the organism on a single source of *external stimuli* by establishing a corresponding focus in sensory cortices, this parallel process may serve to focus upon certain *internal* systems or areas within the cerebral cortex that are required for a given *task*, in effect partially inactivating other, irrelevant systems. William James was the first to our knowledge to conceive of this phenomenon of "internal attention," whereby some mechanism may serve to selectively engage certain regions of the cortex necessary for a particular behavior, while maintaining other regions in a relatively disengaged state. James stated, "Everyone

knows what attention is. It is the taking possession of the mind, in clear and vivid form, of one out of what seem several simultaneous possible objects or trains of thought. Focalization, concentration of consciousness, are of its essence. It implies withdrawal from some things in order to deal more effectively with others. . . ." (James, 1890). Crick (1984) has invoked the concept of an "attentional" searchlight. Luria (1984) was probably the first to suggest a frontal role in orchestrating cortical processes. Fuster (1991) has proposed cortical networks that by virtue of their prefrontal components mediate cross-temporal contingencies. More recently Goldman-Rakic (1990) has used the term "working memory" for this same fundamental concept. Others (Cooper et al., 1968; Samra et al., 1969) have suggested that the dominant thalamus may direct activation important for the modulation and integration of language. Such activation was believed to be more specific than tonic arousal. However, Ojemann (1983) carried this reasoning one step further, suggesting that "thalamic alerting processes selectively activate the specific discrete cortical areas (mosaics) appropriate to the particular language processing" (p. 203). He even suggested that the nonspecific cortical afferents from the intralaminar nuclei might provide the mechanism.

We suggest the term *selective engagement* to define this process. This term conveys the concept of selective, temporary enlistment of specific neural nets to carry out a particular behavior or to maintain a particular mental state, without invoking, possibly inappropriately, such physiologic terms as activation. We consign to attention its traditional meaning of focus upon a particular, external locus and type of sensory input, and avoid the complex historic associations implicit in such terms as working memory. However, selective engagement can be viewed, as Goldman-Rakic (1990) has shown, as the physiologic basis for working memory, that is the means by which not only a cognitive apparatus can be maintained "on line," but also as the means by which recently acquired information pertinent to task performance can be maintained in an active "buffer."

We conclude that thalamic aphasia might plausibly arise directly from lesions of only one nuclear complex, pulvinar-LP, because this is the only region of the thalamus connected with language areas of the cortex. Empirical evidence touching on this concept is limited and mixed. We suggest that much more commonly, thalamic aphasia results from a lesion of the dominant frontal-ITP-NR-CM system, because such a lesion has the effect of preventing selective focal enhancement of thalamocortical transmission in much of the hemithalamus, pulvinar-LP included. Whether pulvinar-LP-cortical transmission, or direct CM-cortical transmission ("nonspecific" cortical afferents) (Bentivoglio, Balercia, & Kruger, 1991) is the critical factor remains to be determined. We are not aware of any studies of the impact of frontal lobe lesions or attentional set on the function of nonsensory relay thalamic nuclei.

IMPACT OF THE THALAMIC SELECTIVE ENGAGEMENT ON CORTICALLY MEDIATED BEHAVIOR

We have suggested that the thalamus plays a role in the selective engagement of cerebral cortex. A question immediately follows: does the thalamic selective engagement mechanism influence any particular type of cortical process, or is its influence relatively nonspecific? Our analysis of language, in conjunction with data on memory dysfunction with thalamic lesions, suggests some general principles that may apply to the role of the thalamic selective engagement mechanism in cognition and behavior. Thalamic lesions **disrupt declarative memory and lexical-semantic aspects of language** (Crosson, 1992), but **seem to spare implicit memory** (Malamut et al., 1992), repetition (phonologic processing), and syntax (Raymer et al., 1997). We suggest therefore that the **declarative/nondeclarative dichotomy** (Squire, 1992) most economically defines which neurobehavioral processes do and do not involve the nuclei of the dorsal tier of the thalamus (DM, pulvinar, LP, anterior nuclear group).

Declarative or explicit memory consists of knowledge of facts and events and is ordinarily available to conscious recollection. It is represented in association cortices throughout the brain. New declarative memories are encoded by a system composed of the hippocampus, parahippocampal, entorhinal and perirhinal cortices, posteromedial orbitofrontal cortex, the medial septal nuclei, the mammillary bodies, and medial portions of the thalamus, particularly the anterior nuclear group and the dorsomedial nucleus (Squire, 1987, 1992). Nondeclarative or implicit memory consists of a heterogeneous collection of abilities including skills and habits, priming, and some forms of classical conditioning. Nondeclarative memory is reflected only in behavioral change, as when one's tennis game improves with practice, or as when an experimental subject demonstrates an autonomic response or a correct decision when reexposed to a stimulus previously presented too briefly to provide even a sense of familiarity, let alone a basis for conscious recognition. Skills, habits, and some forms of classical conditioning are represented in motor and premotor cortex, cerebellum, and possibly the basal ganglia. Priming appears to reflect consciously inaccessible knowledge represented in lower level sensory association cortices. Unlike declarative memory, nondeclarative memory does not appear to require a special auxiliary processing mechanism like the hippocampal system to be instantiated. Patients with impaired declarative memory generally have preserved nondeclarative memory.

Our suggestion that the dorsal tier of the thalamus is involved in declarative processes is a tentative one because of the paucity of data upon which to judge it and because of the existence of phenomena in patients with thalamic lesions, most notably neologisms, that at least superficially appear to violate the principle.

Neologisms: The apparent violation. Paraphasias in patients with thalamic lesions are usually semantic, but phonemic paraphasic errors and neologisms do occur (e.g., Robin & Schienberg, 1990). The conversion of morphemes into strings of phonemes during spontaneous language would seem to represent a skill and therefore a nondeclarative memory. We have argued elsewhere that the production of grammatic morphology is likely to invoke simple associations between morphemes, both grammatic and nongrammatic, that are based exclusively on their positional relationships to each other within a sentence and are learned in the course of language ontogenesis (Nadeau & Gonzalez-Rothi, 1993). This knowledge of morphemic use within a sentence context thus might also constitute a form of skill. If this line of reasoning is correct, then how is it possible for phonemic paraphasias and neologisms to occur in thalamic aphasia, in which we have posited that procedural memory is preserved? There are at least two possible answers. First, there is evidence that the process of becoming literate is associated with the development of awareness of the segmental nature of words, most particularly at the phonological level, but also at the morphemic and syllabic levels (Bertelson & De Gelder, 1989). Thus, the process of becoming literate seems to define *additionally* phonemes, inflectional grammatic morphemes, derivational morphemes (Semenza, Butterworth, Panzeri, & Ferreri, 1990), and syllables as lexical elements in their own right, consciously retrievable like any item in declarative memory, as opposed to relegating them to existence only as intrinsic components of phonologic motoric sequences, grammatic sequences, or words, which, as elements of skill memory, are not consciously retrievable. Other lines of evidence provide additional support for the concept that sublexical components exist in declarative memory, either as an undifferentiated part of the lexicon, or in a sort of sublexicon (Buckingham, 1987). Aphasia studies suggest that morphemes are to some degree represented separately from the rules of combination (Nadeau et al., 1993; Semenza et al., 1990). In slip of the tongue studies, slips involving words stems, affixes, syllables, and rhymes are common (see Dell, 1986; Caplan, 1993). Because these various sublexical elements have become part of the lexicon, they may be inappropriately selected by a semantic–lexical network interface that is deranged by the thalamic lesion and is no longer able to use meaning to reliably select one and only one item from the lexicon. If this is the case, then illiterate patients with thalamic lesions should not make phonemic paraphasic or neologistic errors. On the other hand, patients with cortical lesions might make such errors either because of improperly regulated lexical selection from declarative memory (literate), or because of breakdown in the cortical mechanisms that regulate the skills of phonemic and morphemic sequencing (literate or illiterate). In any event, the neologisms of patients with thalamic lesions should obey the rules for morphemic combination, which are presumably instantiated in the cortex. In every language there are innumerable words that in principle could be generated by the use of ac-

cepted rules of morphemic combination, but are not accepted as real words, for example, curious and curiosity, but not devious and deviosity (Semenza et al., 1990).

Second, it is possible that thalamic lesions so disrupt mutually inhibitory relationships within the semantic field and the lexicon that two competing lexical entrees can be equally activated at a given time (in parallel distributed processing terms (McClelland, Rumelhart, & PDP Research Group, 1986). The simultaneous activation of two lexical entrees may in turn lead to the attempted recruitment of blends of their respective morphemic or phonologic representations, leading to the production of neologisms. Butterworth (1979) found that neologisms are preceded by pauses significantly more often than are normal nouns or semantic paraphasias and that strings of neologisms often do not reveal any evidence of *conduite d'approche*. These observations suggest a problem with lexical engagement that leads to a delay in achieving threshold for lexical output and results in language production that although sensed to be wrong, is not tightly constrained by a single lexical target during attempts at correction, and might be influenced by a number of lexical targets.

Thalamic effect on cortical processing. The thalamic effect on cortical processing may be to transiently and very briefly strengthen connections between subsets of cortical neurons (von der Malsburg in Crick, 1984). This may create momentary, tailor-made, cortical networks that in the realm of language would support the lexical-semantic associations required at that particular moment; in this role, the thalamus would be one of the mechanisms involved in defining working memory. These networks would be expected to exhibit the properties, including graceful degradation, that stem from the distributed representations and massive parallelism envisioned by McClelland and Rumelhart (McClelland et al., 1986). Graceful degradation refers to the relative preservation, as a network is damaged or rendered noisy, of the information most redundantly represented in the strength of network connections. In language, the effect of more lateral thalamic lesions (involving the frontal-ITP-NR-CM system) may be to degrade the cortical neural nets subserving semantic-lexical association, such that the least well represented associations (e.g., low frequency (Raymer et al., 1997)) are selectively unavailable.

Thalamic lesions may also produce dysfunction at the prelinguistic level, i.e., at the level of concept generation. Even though some studies of recall of specific items have suggested that remote memory may be spared in patients with paramedian infarction (Malamut et al., 1992; but see Stuss et al., 1988), Goldenberg et al. (1983) noted disproportionately impaired ability to elaborate in the spontaneous description of remote memories. This is very reminiscent of adynamic aphasia (Costello & Warrington, 1989; Luria, 1984), in which patients with dominant frontal lesions are capable of very limited spontaneous language (when they must generate the concepts them-

selves), but do much better with picture description, in which case the concepts follow more automatically from the visual images. Lhermitte (1984) observed evidence of problems with conceptual delineation and sequencing in patients with thalamic lesions. He noted that "narration is no longer chronologic . . . a passage from one story slides into another story . . . the patient cannot control the order of his ideas." He also noted that disorders of this type are seen in patients with left frontal lesions. We hypothesize that dysfunction of the frontal-ITP-NR-DM system may be responsible, because DM is the major source of thalamic projections to the frontal lobes, and because the frontal-ITP-NR-DM pathway is likely to pass in the midline immediately below the thalamus (Jones, 1975; Steriade et al., 1984), where it would be susceptible to damage by paramedian infarcts, as in the case of Goldenberg et al. (1983). The effect of such lesions may be to degrade the frontal cortical neural nets subserving semantic conceptual development. Just as lexical-semantic impairment with more lateral lesions is frequency dependent, consistent with graceful degradation of distributed representations, so too would graceful degradation at the conceptual level be expected to disproportionately affect detail and to blur conceptual distinctions.

CONCLUSION

We have concluded that aphasia due to nonthalamic subcortical lesions results from a combination of thalamic disconnection and transient or permanent cortical dysfunction related to persistent ischemia or infarction that is not evident on CT or MRI scans. We have also concluded that the basal ganglia have very little to do with language function, a conclusion also reached by Alexander (1992). The evidence favoring these conclusions seems very strong, but if any doubt remains, it centers on Cd_H and in turn on the prefrontal cortex with which it is predominantly connected. The ideal lesion for study is the lacunar infarct of Cd_H. To date, only dysarthria without aphasia has been documented in such patients (Caplan et al., 1990; Mendez et al., 1989; Richfield et al., 1987; Schiff et al., 1983). However, more extensive study might prove rewarding. A second, somewhat less optimal population in which to address this question is Parkinsonian patients. However, to draw useful conclusions, repeated measures will have to be made on and off medication (Delis, Diefenfeld, Alexander, & Kaplan, 1982; Gotham, Brown, & Marsden, 1988; Mohr, Fabbrini, Ruggiero, Fedio, & Chase, 1987; Pillon, Dubois, Bonnet, Esteguy, Guimaraes, Vigouret, Lhermitte, & Agid, 1989a; Pillon, Dubois, Cusimano, Bonnet, Lhermitte, & Agid, 1989b; Pullman, Watts, Juncos, Chase, & Sanes, 1988), because pathological studies have demonstrated that many Parkinsonian patients with dementia, even though they have no evidence of Alzheimer's disease, do have widespread cortical and subcortical pathologic findings that could affect intellectual function (de la Monte, Wells, Hedley-Whyte, & Growdon, 1989). In particu-

lar, such pathology might account for impairment of syntax reported in many patients with Parkinson's disease (Grossman, Carvell, Gollomp, Stern, Vernon, & Hurtig, 1991), whereas syntax was spared in our patient with a VA lesion (Raymer et al., 1992).

Marsden and colleagues (1987) have provided compelling evidence that sequential movements are preferentially involved in Parkinson's disease, and that what is most prolonged is not movement time but intermovement latency. Presumably this reflects the effect of dopamine deficiency in the putamen. If we assume that the grist of Cd_H is concepts, by virtue of its extensive connections with prefrontal cortex, then one result of dopamine deficiency in Cd_H might be an increase in interconcept latency, i.e., bradyphrenia. However, with one exception (Zimmerman, Sprengelmeyer, Fimm, & Wallesch, 1992) experimental studies in Parkinsonian patients have not shown that bradyphrenia (or indeed any cognitive impairment) is sensitive to dopamine replacement (Gotham et al., 1988; Pillon et al., 1989a; Pullman et al., 1988; Rogers, Lees, Smith, Trimble, & Stern, 1987). This may be because dopamine deficiency in Cd_H in Parkinson's disease is considerably less severe than in the putamen (Gotham et al., 1988; Kish, Shannack, & Hornykiewicz, 1988) and therefore generally not sufficiently great to induce Cd_H dysfunction. However, it may also be because Cd_H dopamine deficiency does not induce bradyphrenia, or because bradyphrenia is due to primary cortical pathology (de la Monte et al., 1989). On the other hand, there is recent evidence favoring the notion that dopamine-related impairment in basal ganglia function may have an impact on cognitive processes. In the same way that impairment in the performance of *simultaneous* movements in Parkinson's disease is dopamine responsive (Marsden, 1987), impairment in the performance of simultaneous cognitive tasks can be improved with dopaminergic treatment (Malapani, Pillon, Dubois, & Agid, 1994).

We have concluded in the second major section of this paper that aphasia in patients with dominant thalamic lesions is usually related to destruction of the frontal-ITP-NR-CM system, which may produce linguistic dysfunction by impairing selective disinhibition of thalamocortical neurons in LP and pulvinar or by altering CM transmission to cortex. The result, whichever the mechanism, is defective selection of neuronal nets in the projection field of LP and pulvinar, in effect, a failure to properly define working memory in the temporal and parietal cortex that serves as the substrate for the lexical-semantic field. We have termed this a defect of selective engagement. We have further suggested that the essential province of the selective engagement mechanism supported by the dorsal tier of thalamic nuclei is the processing of material in declarative memory. The most critical tests of these hypotheses will be much more detailed definition of the anatomy of lesions in well-studied patients with aphasia due to thalamic infarctions, better characterization of linguistic behavior in such patients, further attempts to correlate types of linguistic and prelinguistic deficits with lesion locus in the thala-

mus, and the careful study of deficits in nonlinguistic processes caused by thalamic lesions. We also hope that with the recent emergence of a relatively detailed understanding of the effect of arousal on gated relay in the thalamus (Steriade et al., 1990), the focus of neurophysiologic research will shift to the role of the thalamus in attention and selective engagement and, more particularly, the mechanisms of the frontal-ITP-NR-CM system.

APPENDIX

LOCATION, CONNECTIVITY, AND VASCULAR SUPPLY OF MAJOR THALAMIC NUCLEI

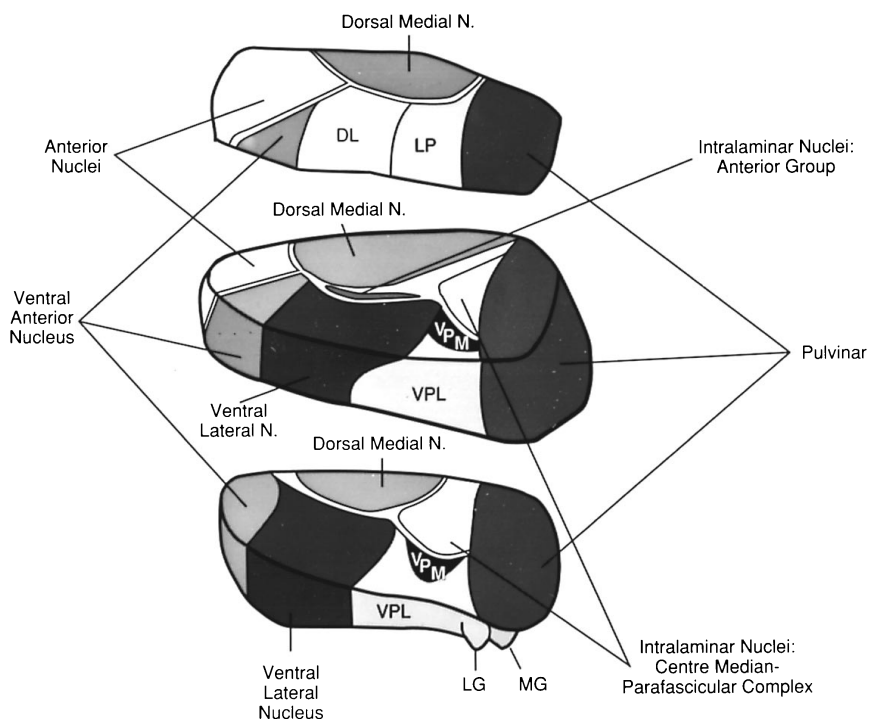


FIG. 8. Sketch of the thalamus. DL, dorsolateral (also commonly referred to as lateral dorsal); LG, lateral geniculate; MG, medial geniculate; VPL, ventral posterolateral; VPM, ventral posteromedial. Drawings are composed to indicate relationships between structures but are not meant to be anatomically precise. (From Crosson, B. *Subcortical functions in language and memory*, Fig. 1.5, p. 24. New York: Guilford Press, 1992; with permission).

TABLE 5
Summary of the Major Thalamic Nuclei, Their Connections, and Vascular Supply

	Afferent sources	Efferent targets	Vascular supply
VPL, VPM	Medial lemniscus, spinothalamic tract, principal and spinal nuclei of cranial nerve 5	SI (3,1,2) SII	Thalamogeniculate
VLP	Cerebellum, spinothalamic tract vestibular nuclei	Precentral gyrus (4)	Tuberothalamic, thalamogeniculate
VLa	GP _i , SN _{PR}	SMA, 4, 6	Tuberothalamic
VA	GP _i , SN _{PR}	Prefrontal and inferior parietal cortex, 6	Tuberothalamic
MG	Inferior colliculus	AI, AII, and neighboring auditory cortices	Posterior choroidal
LG	Retina via optic tract, superior colliculus, pretectum	17	Anterior and posterior choroidal
LP		7	Posterior choroidal
Pulvinar			
Medial		Prefrontal cortex, 22, lateral nucleus of amygdala	Posterior choroidal, thalamogeniculate (minimal)
Anterior		5	Same
Lateral, inferior	Superior colliculus, pretectum	7a, striate and prestriate (lateral and inferior temporal, occipital cortex), angular and supramarginal gyri	Same
Intralaminar			
Rostral (central medial, central lateral, paracentral, rhomboid)	Prefrontal, limbic, parietal cortex, 4, 6, spinothalamic tract, cerebellum	Striatum, nucleus accumbens, widespread cerebral cortex	Paramedian

Caudal (CM, parafascicularis)	Prefrontal cortex, 4, 6, cerebellum, striatum, GP _i , midbrain and pontine reticular formation, deep superior colliculus, preectum, vestibular nuclei	Striatum, widespread cortex	cerebral	Paramedian
Reticular nucleus	Cerebral cortex (especially frontal via ITP), midbrain and pontine reticular formation, locus ceruleus (norepinephrine), raphe nuclei (serotonin)	All thalamic nuclei, particularly intralaminar, except anterior nuclear group		Various
DM				
Paralamellar	SN _{pr} , intralaminar nuclei, superior colliculus, reticular formation, preectum, spinothalamic tract, central gray	Frontal eye fields		Posterior choroidal, tuberothalamic (minimal), paramedian (inferior)
Parvocellular	Intralaminar nuclei, ventral tegmental area (dopamine)	Dorsolateral prefrontal		Same
Magnocellular	Olfactory tubercle, piriform and prepiriform cortex, amygdala, temporal neocortex, GP _i , substantia innominata	Orbitofrontal cortex, insula		Same
Anterior nuclear group, DL	Hippocampus (via subiculum, pre-, and parasubiculum), mammillary bodies, locus ceruleus	Cingulate gyrus, retrosplenial area, pre- and parasubiculum		Tuberothalamic, posterior choro- roidal

Note. With the exceptions of the reticular nucleus (NR) and the intralaminar nuclei, efferent connections to the cerebral cortex are uniformly reciprocated by topographically organized afferent corticothalamic projections. Numbers refer to Brodmann's areas. The thalamogeniculate and posterior choroidal arteries originate from the proximal posterior cerebral arteries, the tuberothalamic from the posterior communicating artery, the paramedian from the basilar artery tip or the proximal posterior cerebral artery, and the anterior choroidal from the supraclinoid carotid artery. VP_L, ventral posterolateral nucleus; VP_M, ventral posteromedial nucleus; VL_{pr}, ventrolateral nucleus, posterior part; VL_{sa}, ventrolateral nucleus, anterior part, VA, ventral anterior nucleus; MG, medial geniculate; LG, lateral geniculate; LP, lateral posterior; CM, center median; DM, dorsomedial; DL, lateral dorsal; SI, primary somatosensory area (3,1,2); SII, secondary somatosensory area; SMA, supplementary motor area; AI, primary auditory cortex; AII, secondary auditory cortex; GP_i, globus pallidus, internal part; SN_{pr}, substantia nigra pars reticulata; ITP, inferior thalamic peduncle.

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